

IoT-Based Two-Way Sign Language Understanding & Response Hub

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SELECTED THEME: ASSISTIVE TECHNOLOGY / IOT & ARTIFICIAL INTELLIGENCE



Description of the identified problem: There is a significant communication barrier between the deaf/hard-of-hearing community and individuals who do not understand sign language. Current translation solutions often require expensive, specialized hardware or heavy cloud computing which introduces latency.

Importance and relevance of the problem: Effective and instantaneous communication is a fundamental human right. Real-time translation is critical for healthcare, education, and daily accessibility.

Target users or beneficiaries: -

- Deaf and hard-of-hearing individuals.
- Public service workers, teachers, and medical staff.
- IoT researchers looking for cost-effective edge-computing solutions.

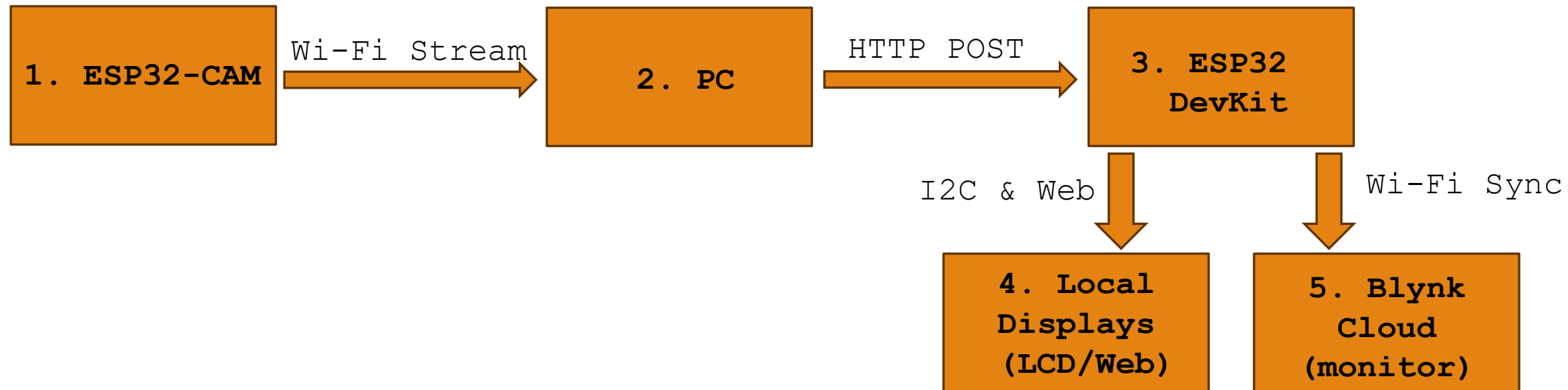
Overview of the proposed solution: A distributed IoT architecture that offloads heavy AI processing to a local PC, while utilizing low-cost microcontrollers for capture and user interfaces.

Key functionalities:

- Real-time hand gesture tracking and classification using MediaPipe.
- Multi-modal output: Displays translated text simultaneously on a physical LCD and a hosted Web HUD.
- Remote system monitoring via the Blynk IoT Cloud platform.

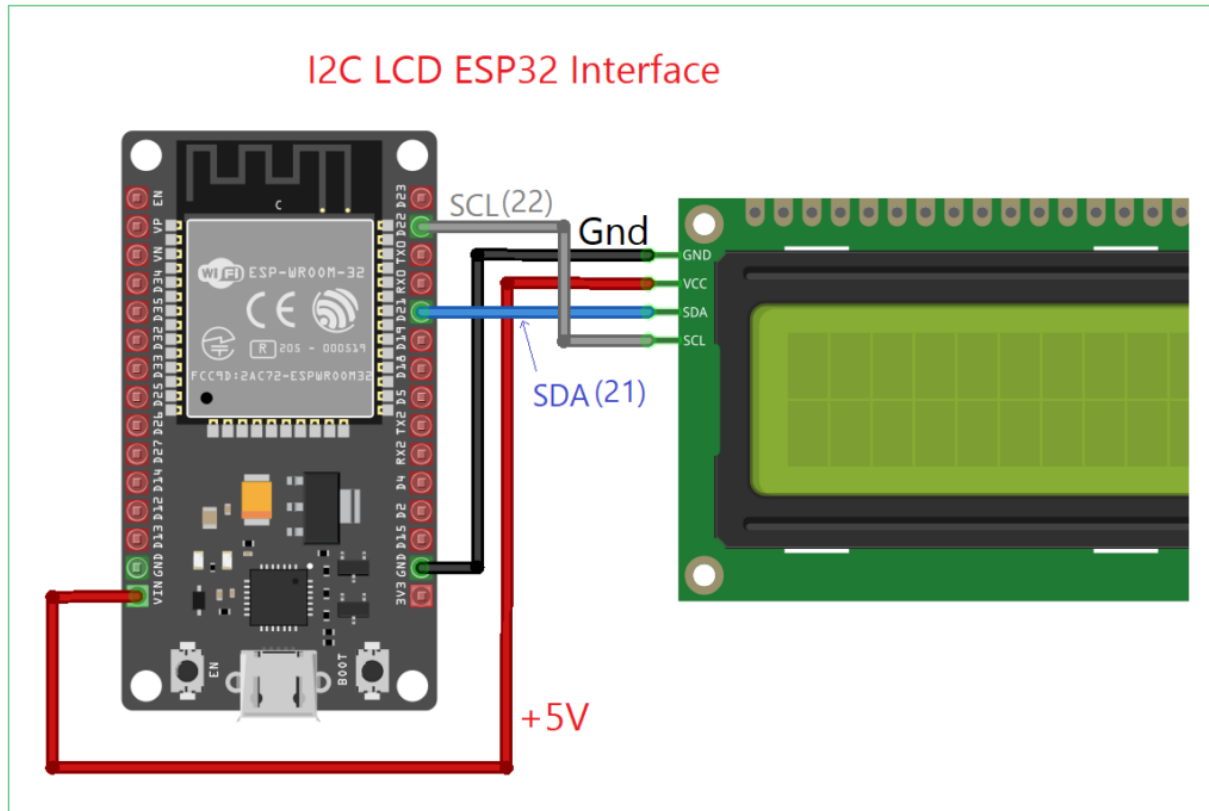
Expected impact: Provides a low-latency, highly accessible translation hub that can be deployed in a room using off-the-shelf components, bridging the communication gap seamlessly.

System Architecture / Block Diagram



Circuit Diagram

Microcontroller used: ESP32 DevKit (Main Hub) & ESP32-CAM (Vision Node)



Hardware connections (ESP32 DevKit to LCD): -

- **GND ESP32 GNDVCC**
- **ESP32 VIN** (5V required for backlight brightness)
- **SDA ESP32 GPIO 21**
- **SCL ESP32 GPIO 22**



**Esp 32 cam
combined with
Esp 32 cam mb**

Hardware & Software Requirements

Hardware:

ESP32 Development Board (Wi-Fi/Server capabilities)

ESP32-CAM Module (For remote video capture)

16x2 I2C LCD Display

PC/Laptop (e.g., Intel Core i7 with GPU for inference processing)

Software:

Programming Languages: Python 3 (AI Inference), C++ (Microcontroller logic),
HTML/Tailwind CSS/JS (Web UI).

Frameworks/Libraries: Arduino IDE, LiquidCrystal_I2C

AI/ML Models: Google MediaPipe Tasks API (Hand Landmarker Model), OpenCV.

Cloud Services: Blynk IoT Platform

Limitation

Current constraints: - The system heavily relies on a stable local Wi-Fi network; network congestion can introduce latency between the AI detection and the ESP32 output.

Lighting conditions and background clutter can affect the camera's ability to accurately detect hand landmarks.

Assumptions made: - The user signing is relatively stationary and facing the camera.

Technical challenges: - Balancing frame rate vs. resolution on the ESP32-CAM to ensure the PC AI gets a clear but fast enough image.

Known limitations: - The current AI logic uses static geometric rules (e.g., finger bending), which only supports static signs (like numbers/alphabets) rather than dynamic, moving signs.

Future Expansion

Additional features: - Implement a physical output using a Robotic Hand (utilizing servos and flex sensors) to respond back in physical sign language.

Transition from static sign detection to a Recurrent Neural Network (RNN) or LSTM model to translate continuous, dynamic sign language sequences.

Scalability possibilities: - Upgrading the local PC to an edge AI device like a Raspberry Pi 4 to make the entire system compact and portable.

Long-term impact: - Creating a fully autonomous, two-way communication kiosk for public spaces (banks, hospitals) that requires zero human intervention to facilitate conversations.